

Multiple Choice (10 marks)

1. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
2. Which one of the following is equal to $P(A, B, C)$ given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?
 - (a) $P(A | B) * P(B | C) * P(C | A)$
 - (b) $P(C | A, B) * P(A) * P(B)$
 - (c) $P(A, B | C) * P(C)$
 - (d) $P(A | B, C) * P(B | A, C) * P(C | A, B)$
3. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Consider the Bayesian network given below. How many independent parameters are needed for this Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 2
 - (b) 4
 - (c) 8
 - (d) 16
5. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 1, 1, 1 \\ 1, 1, 1 \\ 1, 1, 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3

True / False (10 marks)

Answer True or False

6. True or False: Both the original ResNets and Transformers are feedforward neural networks designed for different tasks in deep learning.
7. True or False: Predicting the amount of rainfall in a region based on various cues is a supervised learning problem.
8. True or False: The joint probability $P(H, U, P, W)$ is calculated as $P(H) * P(U) * P(P | H) * P(W | P)$.
9. True or False: Random crop and vertical flip are the most effective image data augmentation techniques for enhancing natural images.
10. True or False: Increasing the amount of training data is ineffective in combating model overfitting.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to remove leading zeroes from an ip address.
12. Write a function to remove all elements from a given list present in another list.

End of Paper